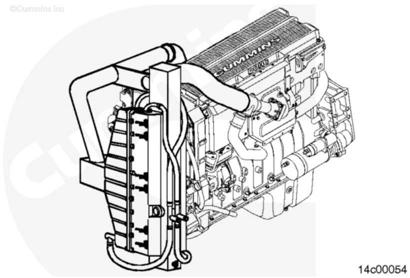
Trainings

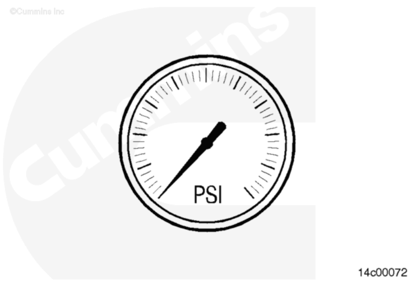
Run-In Instructions

Follow the general operating procedures and safety precautions when performing the engine run-in. Refer to Procedure 014-005 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-005-tr.html)



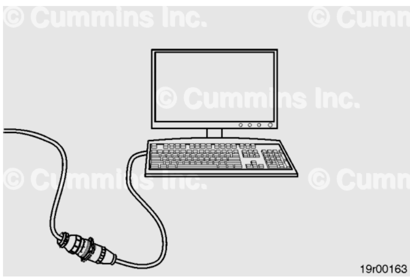
Use the Dynamometer Worksheet to record all of the engine performance parameters. Refer to Procedure 014-001 in Section 14. (/qs3/pubsys2/xml/en/procedures/101/101-014-001-tr.html)

Cummins Inc. recommends monitoring block coolant pressure during run-in to aid in early indication of a cooling system problem.



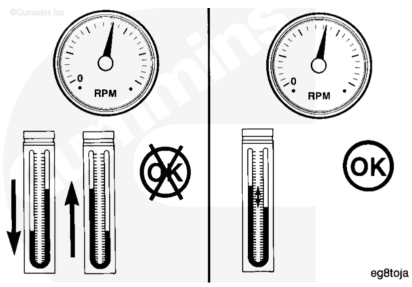
Note : INSITE™ electronic service tool must be used to confirm that the engine control module (ECM) code on the ECM dataplate is the same as the one installed in the ECM. There is an associated fuel pump code with all ECM codes.

Use INSITE™ electronic service tool to obtain the ECM code and the fuel pump code from the ECM dataplate.



During the test, if a sudden increase in blowby occurs or if blowby exceeds the maximum allowable limit during any run-in, return to the previous step of the test and continue the run-in. If blowby does **not** reach an acceptable level during the next step, discontinue the run-in and determine the cause.

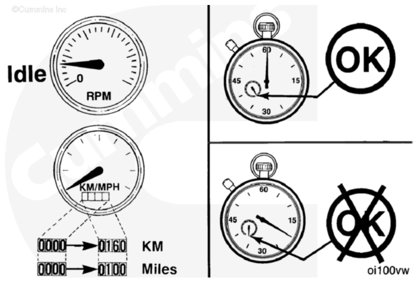
Do **not** proceed to the next step until a steady, acceptable blowby reading is obtained.



Note : Blowby must be measured by the blowby checking tool, Part Number 3822566 (ISM) or Part

Number 3822567 (ISX), and water manometer, Part Number ST-1111-3.

Do **not** idle the engine for more than 5 minutes at any one time during the first 3 hours or 160 km [100 mi] of operation.



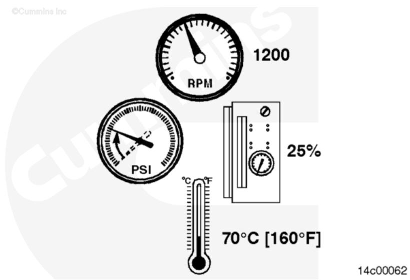
Adjust the engine speed to 1200 rpm. Apply a test load sufficient to develop 25 percent of advertised horsepower of the engine.

Operate the engine at this speed and load level until the coolant temperature is 71°C [160°F].

Check for leaks. Fix all leaks.

Check all gauges and record the data.

Do **not** proceed to the next step until the blowby becomes stable within specifications.

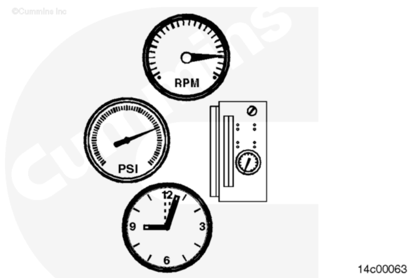


Open the throttle to obtain the engine speed at which advertised horsepower is developed, and adjust the dynamometer load to achieve 75 percent of advertised horsepower on the engine.

Operate the engine at this speed and load level for 3 minutes.

Check all gauges and record the data.

Do **not** proceed to the next step until the blowby becomes stable within specifications.

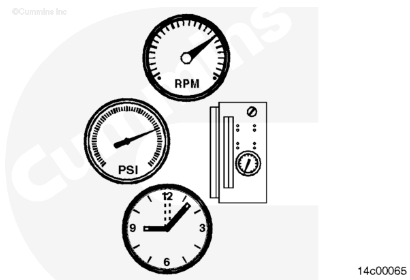


Maintain the engine speed at which advertised horsepower is developed. Move the throttle lever to its fully opened position, and increase the dynamometer load until 100 percent advertised horsepower of the engine is developed.

Operate the engine at this speed and load level for 5 minutes.

Check all gauges and record the data.

Do **not** proceed to the next step until the blowby becomes stable within specifications.

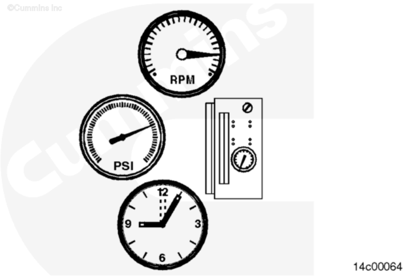


Increase the dynamometer load until the engine speed reduces to the engine's peak torque rpm.

Operate the engine at this speed and load level for 7 minutes.

Check all gauges and record the data.

Do **not** proceed to the next step until the blowby becomes stable within specifications.

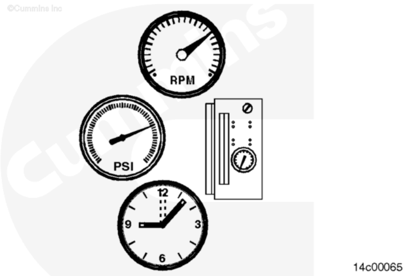


Reduce the dynamometer load until the engine speed increases to the engine speed at which advertised horsepower is developed.

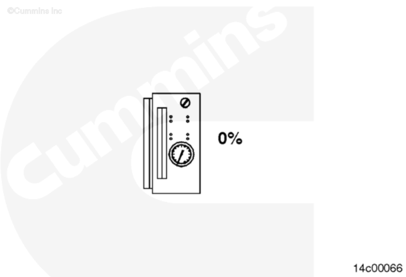
Operate the engine at this speed and load level for 5 minutes.

Check all gauges and record the data.

Do **not** proceed to the next step until the blowby becomes stable within specifications.

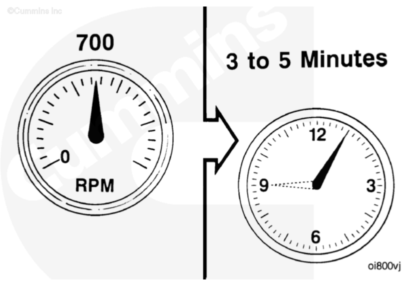


Remove the dynamometer load completely.



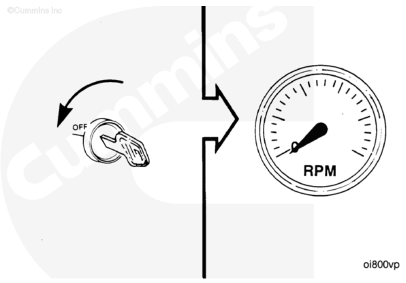
⚠ CAUTION ⚠

Do not shut off the engine immediately after the run-in is completed. Allow the engine to cool by operating it at 700 to 900 rpm for a minimum of 3 to 5 minutes to avoid internal damage. This allows the turbocharger and other components to cool.



Shut the engine off.

Make sure all instrumentation is removed before removing the engine from the engine dynamometer.



Last Modified: 22-Aug-2012